Data Analytics Internship Assessment

Cloud Kitchen Market Intelligence Project

Submission Deadline

48 Hours from Assignment Receipt

Objective

The purpose of this assessment is to evaluate your ability to:

- Collect structured data from real-world platforms
- Investigate APIs and network requests
- Clean and organize datasets
- Perform business-oriented analysis
- Communicate findings clearly

This is not a coding challenge alone. We are evaluating your analytical process, attention to detail, and ability to generate insights.

Task Overview

You are required to conduct a mini market intelligence study of food delivery restaurants operating in Nashik.

Select any ONE locality from the following:

- College Road
- Gangapur Road
- Indira Nagar
- Dwarka
- Canada Corner
- Panchavati

Part 1: Data Collection

Collect data for a minimum of 30 restaurants from either Swiggy or Zomato.

For each restaurant capture:

- Restaurant Name
- Cuisine(s)
- Rating

- Number of Reviews (if available)
- Cost for Two
- Locality
- Delivery Time
- Whether Restaurant Appears to be:
    - Cloud Kitchen
    - Dine-In Restaurant
    - Unknown

Store data in CSV format.

Part 2: Network Investigation

Using browser developer tools:

1. Inspect network requests while browsing restaurant listings.
2. Identify at least three API or JSON endpoints used by the platform.
3. Document:
    - Request URL pattern
    - Request method
    - Response type
    - Information returned

Required Evidence:

- Screenshots of Network Tab
- Screenshots of JSON Responses
- Short explanation of findings

Important:

We are evaluating your investigative process, not just your final answer.

Part 3: Menu Intelligence

Select 5 restaurants from your dataset.

For each restaurant:

Collect:

- Menu Categories

- Top 10 Menu Items
- Item Prices

Additionally:

Identify:

- Highest Priced Item
- Lowest Priced Item
- Estimated Bestseller Item (explain reasoning)

Create a structured menu dataset.

Part 4: Data Cleaning

Create a cleaned version of your dataset.

Document:

- Duplicate detection strategy
- Missing value handling
- Cuisine normalization approach

Provide:

Before Cleaning:

- Number of Records

After Cleaning:

- Number of Records

Explain every transformation performed.

Part 5: Business Analysis

Using only the data you collected:

Answer the following:

Question 1

Which cuisines appear most saturated in your selected locality?

Support your answer with data.

Question 2

Which cuisine category appears underrepresented?

Explain your reasoning.

Question 3

If you were launching a cloud kitchen in this locality with ₹5 lakh capital, what cuisine would you launch first?

Provide:

- Target cuisine
- Target customer
- Expected pricing strategy
- Competitive advantage

Question 4

Identify one restaurant that appears operationally efficient and explain why.

Use observable evidence.

Part 6: SQL Exercise

Create a database schema for:

- Restaurants
- Menu Items

Write SQL queries for:

1. Top 5 highest rated restaurants
2. Average cost-for-two by cuisine
3. Restaurants with more than one cuisine tag
4. Highest priced menu item

You do not need to execute the queries, but they must be syntactically correct.

Part 7: Methodology Report

Prepare a short report (2–4 pages) describing:

- How data was collected
- Tools used
- Problems encountered

- Assumptions made
- What you would improve if given more time

Submission Format

Submit a ZIP file containing:

1. Raw Dataset
2. Cleaned Dataset
3. Menu Dataset
4. SQL File
5. Methodology Report (PDF)
6. Screenshots Folder
7. Python Scripts / Notebooks (if used)

Important Notes

We are interested in your process and reasoning.

Submissions generated entirely through AI tools without evidence of independent investigation, screenshots, methodology, and reproducible work will be rejected.

You may use AI assistants as a reference tool, but all conclusions must be supported by your own collected data and documented workflow.